

Network Robustness and Resilience: Theory and Applications in Human, Natural, and Infrastructure Systems



Invited Speaker

Jianxi Gao

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Date: October 9, 2026 (Fri)

Time: 8:30am (HKT)

Zoom Meeting: 801 137 0362

Biography

Jianxi Gao is an Associate Professor in the Computer Science Department at Rensselaer Polytechnic Institute. His research uses network theory, control theory, statistical physics, artificial intelligence, and operations research to understand, predict, and ultimately control complex systems' resilience and cascading failures in complex networks. He received the NSF CAREER Award in 2021 to study the resilience of complex networks. He has published over 100 journal papers and over 10 conference with over 11k citations on Google Scholar. Dr. Gao has also been selected as the Editor Board of Scientific Reports and Physica A, an external PNAS editor, and a distinguished EPL/ Elsevier referee.

Abstract

Our world is increasingly fragile: cascading failures in power grids, breakdowns in global supply chains, species extinction in ecosystems, and intensifying climate shocks now routinely disrupt the systems we depend on. Yet many complex systems also surprise us with their ability to recover, reorganize, and even improve after disaster. In this talk, I explore how and why this happens. I will first unpack what “robustness” and “resilience” really means across social science, ecology, and infrastructure systems. I then introduce dimension-reduction approaches that help us predict system resilience and strategically guide network restoration after major disruptions. Drawing on cases from social, ecological, and infrastructural systems, I show how network structure and adaptation together determine whether systems collapse, bounce back, or transform. Finally, I will also introduce how machine learning improve our understanding and inferencing the resilience of complex networks. The talk argues that resilience is not just about surviving shocks, it is about learning to adapt with uncertainty in a rapidly changing world.